EXPERIMENT NO: 20

Experiment to understand K-means clustering algorithm for a given data set.

Program:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv('Mall_Customers.csv')
print(df.info())
print(df.head())
sns.pairplot(df)
features=df.iloc[:,[3,4]].values
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
print(Final.head())
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8).map(plt.scatter,"Annual Income (k$)","Spending Score (1-100)").add_legend();
plt.show()
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
```

```
  model=KMeans(n_clusters=i)
  model.fit(features_el)
  wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

Output:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	200 non-null	int64
1	Gender	200 non-null	object
2	Age	200 non-null	int64
3	Annual Income (k		

$)200non - nullint644SpendingScore(1 - 100)200non - nullint64dtypes: int64(4), object(1)memo$

) Spending Score (1-100)

0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

/tmp/ipykernel_4411/1786908403.py:15: SettingWithCopyWarning:

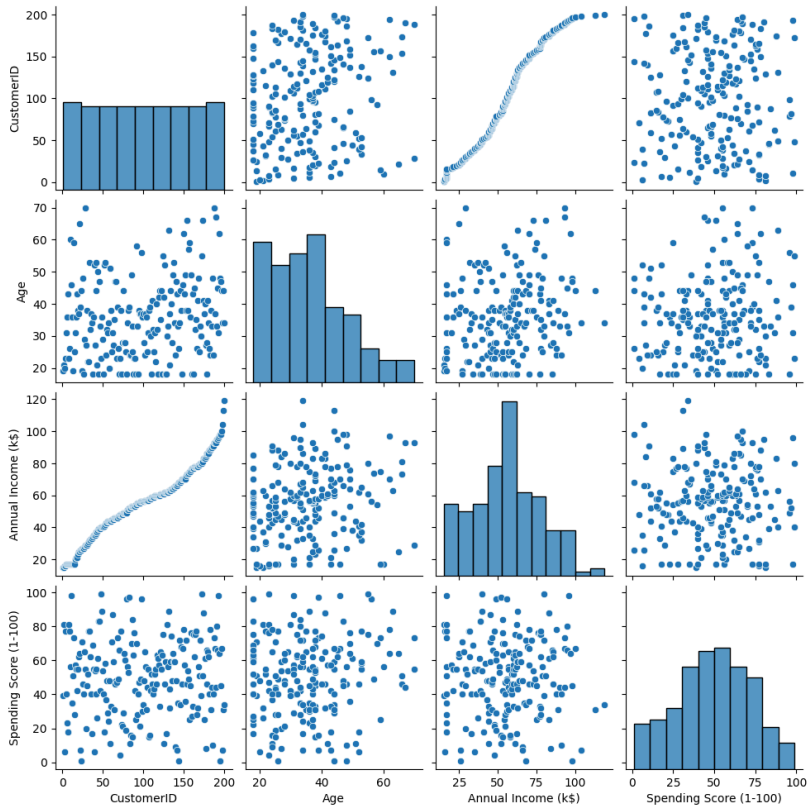
A value is trying to be set on a copy of a slice from a DataFrame.

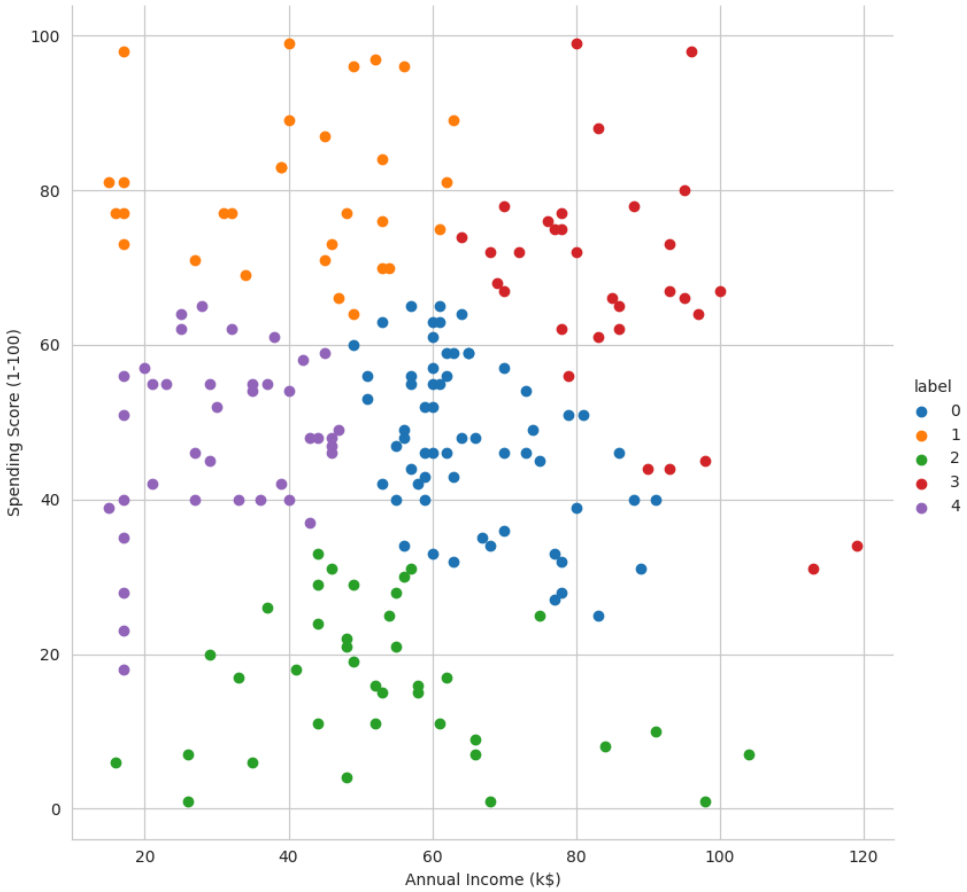
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

Final['label']=model.predict(features)

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	1
2	16	6	2
3	16	77	1
4	17	40	4





[<matplotlib.lines.Line2D at 0x7e90cd081810>]

